

Summary

Let's summary, on one hand we have

$$Z_i \mid (Z_{(1)}, \dots, Z_{(n)}) \sim \frac{1}{n} \sum_{j=1}^n \delta_{Z_{(j)}}$$

On the other hand, we have

$$Z_i \mid \hat{P}_n \sim \hat{P}_n, \quad \hat{P}_n = \frac{1}{n} \sum_{i=1}^n \delta_{Z_i},$$

We also have

$$\frac{1}{n} \sum_{i=1}^n \delta_{Z_i} = \frac{1}{n} \sum_{i=1}^n \delta_{Z_{(i)}}$$

Therefore

$$Z_i \mid (Z_{(1)}, \dots, Z_{(n)}) \stackrel{d}{=} Z_i \mid \hat{P}_n$$

Example

For example, suppose $n = 3$ and the observations, in an unknown order, are

$$\{2, 5, 5\}.$$

Then

$$\widehat{P}_3 = \frac{1}{3}\delta_2 + \frac{2}{3}\delta_5.$$

Conditioning on

$$\widehat{P}_3 = \frac{1}{3}\delta_2 + \frac{2}{3}\delta_5$$

means:

We know the unordered multiset of observations is $\{2, 5, 5\}$, but we do not know which coordinate contains which value.

Under exchangeability, each coordinate is equally likely to occupy each position in that multiset. Therefore,

$$Z_i \mid \widehat{P}_3 \sim \frac{1}{3}\delta_2 + \frac{2}{3}\delta_5,$$

meaning

$$\Pr(Z_i = 2 \mid \widehat{P}_3) = \frac{1}{3}, \quad \Pr(Z_i = 5 \mid \widehat{P}_3) = \frac{2}{3}.$$

Thus, writing " $\widehat{P}_n = 1/n$ " is a type mismatch: $1/n$ is a number, whereas $\widehat{P}_n$ is an entire distribution. Each observation contributes mass $1/n$, but repeated observations accumulate multiple contributions.

$$\varepsilon \sim N(0, \sigma^2)$$

$$y_i = \alpha + \beta x_i + \varepsilon_i \quad i=1, \dots, n$$

Data: $\{(x_i, y_i)\}_{i=1}^n$

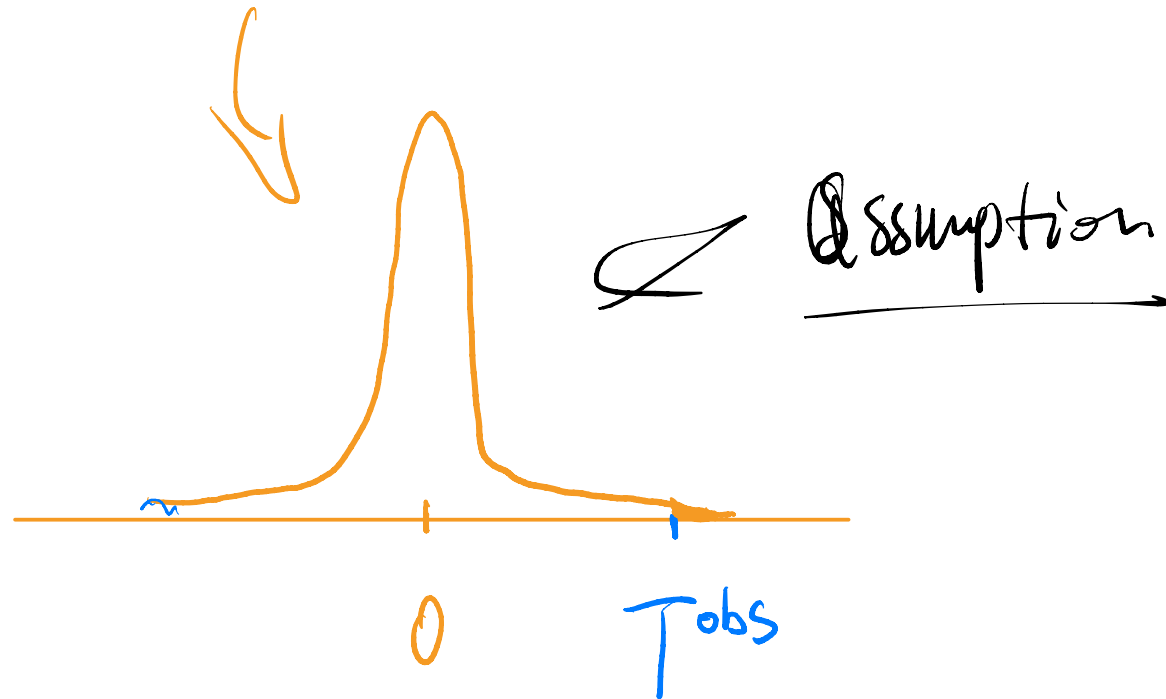
Hypothesis: $H_0: \underline{\beta = 0}$ $H_a: \beta \neq 0$

t-test

$$T = \frac{\hat{\beta} - \underbrace{\beta_{H_0}}_{=0}}{se(\hat{\beta})} = \frac{\hat{\beta}}{se(\hat{\beta})}$$

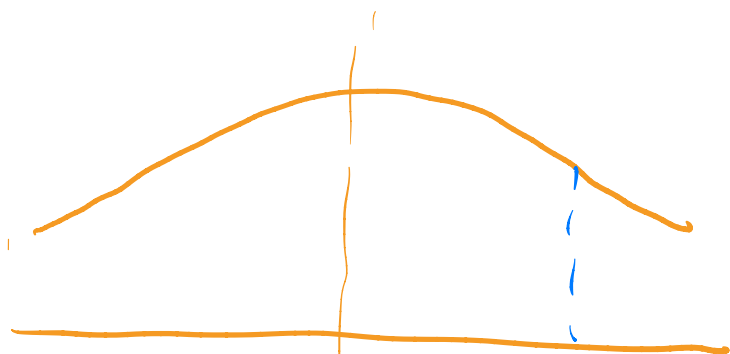
T^{obs} : actual obs, t statistic from $\{(x_i, y_i)\}_{i=1}^n$

$$T \sim t_{n-2} \quad \text{under } \beta = 0$$

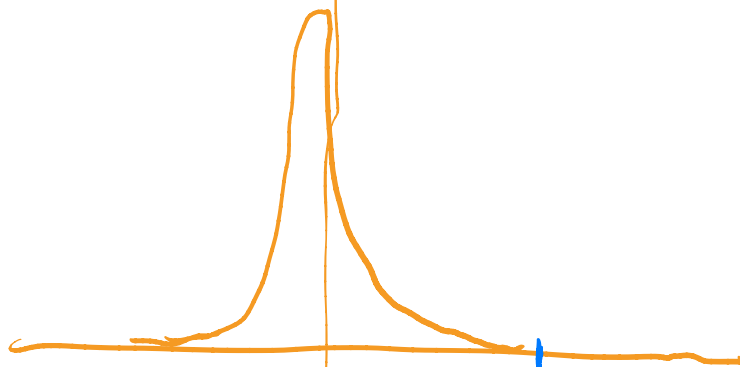


$$P = P(T > T^{\text{obs}} \text{ or } T < -T^{\text{obs}})$$

$$= P(|T| > |T^{\text{obs}}|)$$



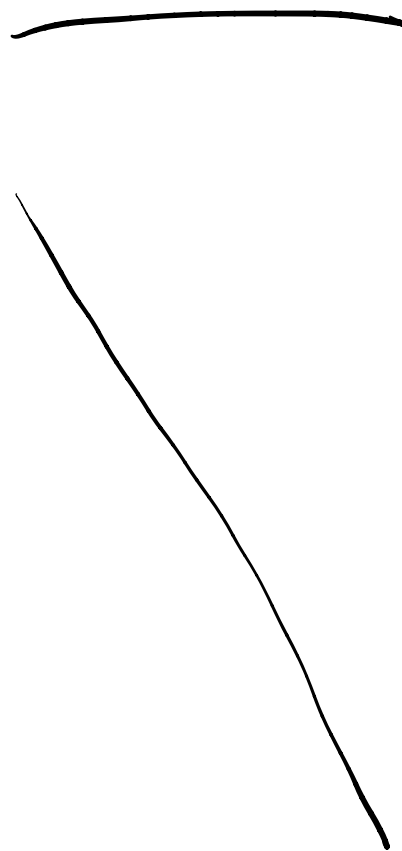
Tobs



Tobs ←

x	y
	y_1
	y_n

under $\beta = 0$



x	y
	$y_{\sigma(1)}$
	$y_{\sigma(n)}$
x	y
	$y_{\sigma(1)}$
	$y_{\sigma(n)}$

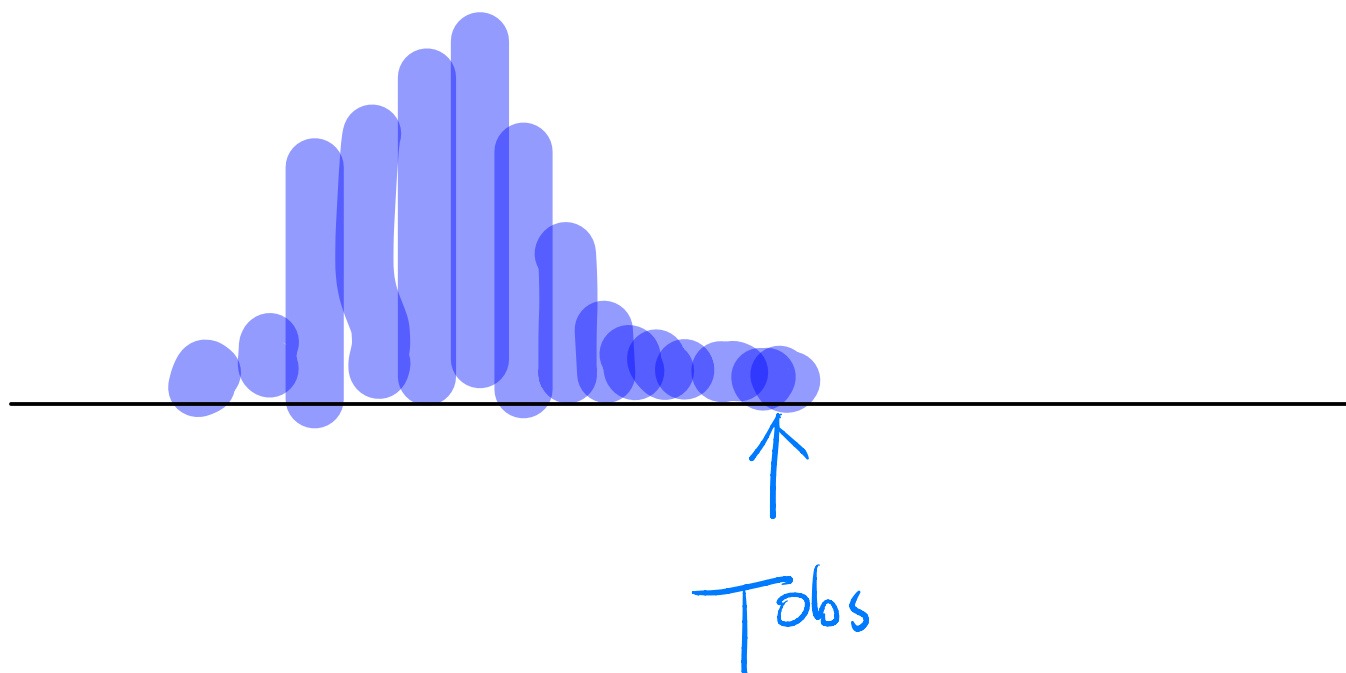
$\rightarrow T^{(1)}$

$\vdots$

$\rightarrow T^{(n)}$

$T^{(1)} \quad T^{(2)} \quad \dots \quad T^{(M)}$

T_{obs}



$$p = \frac{\sum_{i=1}^m \mathbb{1}(|T^{(i)}| \geq |T^{obs}|)}{m}$$

Permutation tests

- Permutation tests are used in statistics for a wide range of different inference tasks.
- permutation tests can be viewed as testing the null hypothesis of exchangeability.

$$Z^n \sim P$$

Let $\mathcal{P}$ be the set of all distributions on $\mathcal{Z}^n$, and let $\mathcal{P}_{\text{exch}} \subseteq \mathcal{P}$ be the subset of distributions for which exchangeability is satisfied. Consider a random vector $(Z_1, \dots, Z_n)$ drawn from some joint distribution P . We would like to perform a hypothesis test of

$$H_0 : P \in \mathcal{P}_{\text{exch}} \quad \text{versus} \quad H_1 : P \in \mathcal{P} \setminus \mathcal{P}_{\text{exch}}.$$

Lets say a large value of our test statistic $T(Z_1, \dots, Z_n)$ will indicate evidence against exchangeability.

$$p = \frac{\sum_{\sigma \in \mathcal{S}_n} \mathbb{1} \{T(Z_{\sigma(1)}, \dots, Z_{\sigma(n)}) \geq T(Z_1, \dots, Z_n)\}}{n!}$$

H_0
 $n!$ total
 σ - permutation
 $\mathcal{S}_n$ all permutation
 $P(p \leq \tau) \leq \tau$

The following theorem shows it is a valid p-value

Theorem 2.4: Validity of the permutation test

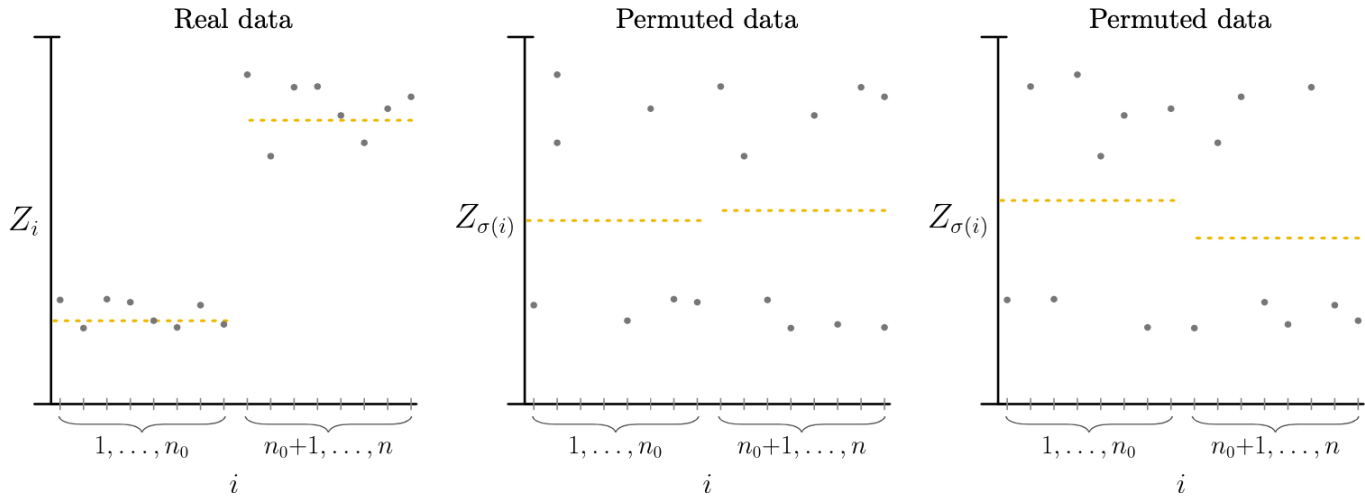
For any function $T : \mathcal{Z}^n \rightarrow \mathbb{R}$, the p-value p defined in (2.3) satisfies $\mathbb{P}_P(p \leq \tau) \leq \tau$ for all $\tau \in [0, 1]$ and all $P \in \mathcal{P}_{\text{exch}}$.

under H_0

it is common to avoid the computational burden of computing all $n!$ permutations by instead sampling a smaller number M of permutations $\sigma_1, \dots, \sigma_M \in \mathcal{S}_n$ **uniformly at random**,

$$p = \frac{1 + \sum_{m=1}^M \mathbb{1} \{T(Z_{\sigma_m(1)}, \dots, Z_{\sigma_m(n)}) \geq T(Z_1, \dots, Z_n)\}}{1 + M}. \quad (2.4)$$

↙
 $n!$



Theorem 2.5: Validity of the permutation test with random permutations

For any function $T : \mathcal{Z}^n \rightarrow \mathbb{R}$, the p-value p defined in (2.4) satisfies $\mathbb{P}_P(p \leq \tau) \leq \tau$ for all $\tau \in [0, 1]$ and all $P \in \mathcal{P}_{\text{exch}}$, where the probability is now taken with respect to both the random draw of $(Z_1, \dots, Z_n) \sim P$, and the permutations $\sigma_1, \dots, \sigma_M$ sampled uniformly at random (with replacement) from $\mathcal{S}_n$.

Example

Testing equality of distributions. Suppose that we have two independent samples from two potentially different distributions, with n_0 draws from P_0 and $n_1 = n - n_0$ draws from P_1 . Without loss of generality, we can take $Z_1, \dots, Z_{n_0} \stackrel{\text{i.i.d.}}{\sim} P_0$ and $Z_{n_0+1}, \dots, Z_n \stackrel{\text{i.i.d.}}{\sim} P_1$. If $P_0 = P_1$, then the Z_i 's are i.i.d. from a single shared distribution $P_0 = P_1$, and therefore exchangeability holds. If we conjecture that any potential difference between P_0 and P_1 would likely lead to a difference of means, we might choose the test statistic

$$T(z_1, \dots, z_n) = \left| \frac{1}{n_0} \sum_{i=1}^{n_0} z_i - \frac{1}{n_1} \sum_{i=n_0+1}^n z_i \right|,$$

$$T(z_1, \dots, z_n) = \sup_{v \in \mathbb{R}} \left| \frac{1}{n_0} \sum_{i=1}^{n_0} \mathbb{1}\{z_i \leq v\} - \frac{1}{n_1} \sum_{i=n_0+1}^n \mathbb{1}\{z_i \leq v\} \right|,$$

Empirical
CDF

$\sim P$
 $(z_1, \dots, z_{n-1}), z_n$

Example

Testing if a new data point is an outlier. Next, suppose that we would like to test whether a particular data point—say, the last data point Z_n —is an outlier relative to the rest of the sequence.

Consider the test statistic

$H_0: Z_n \sim P$

$$T(z_1, \dots, z_n) = \sum_{i=1}^n \mathbb{1}\{z_n > z_i\},$$

the permutation test p-value can be simplified. Observe that

$$T(Z_{\sigma(1)}, \dots, Z_{\sigma(n)}) = \sum_{i=1}^n \mathbb{1}\{Z_{\sigma(n)} > Z_{\sigma(i)}\} = \sum_{i=1}^n \mathbb{1}\{Z_{\sigma(n)} > Z_i\},$$

we can then see that $T(Z_{\sigma(1)}, \dots, Z_{\sigma(n)}) \geq T(Z_1, \dots, Z_n)$ if and only if $Z_{\sigma(n)} \geq Z_n$,

$$p = \frac{\sum_{\sigma \in \mathcal{S}_n} \mathbb{1}\{T(Z_{\sigma(1)}, \dots, Z_{\sigma(n)}) \geq T(Z_1, \dots, Z_n)\}}{n!} = \frac{1}{n!} \sum_{\sigma \in \mathcal{S}_n} \mathbb{1}\{Z_{\sigma(n)} \geq Z_n\}$$

$\nearrow n!$

$$= \frac{1}{n!} \sum_{i=1}^n \underbrace{\left(\sum_{\sigma \in \mathcal{S}_n, \sigma(n)=i} \right)}_{(n-1)!} \mathbb{1}\{Z_i \geq Z_n\} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{Z_i \geq Z_n\}.$$

In fact, this example can be derived in a simpler way. By definition of p , we can verify that, for any $\tau \in [0, 1]$,

$\otimes$ later

$$p = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{Z_i \geq Z_n\} \leq \tau \iff \underbrace{Z_n}_{\text{later}} > Z_{(k)} \text{ for } k = \lceil (1 - \tau)n \rceil.$$

We know the exchangeability of $Z_1, \dots, Z_n$ implies that

$\mathbb{P}(p \leq \tau)$

$$\mathbb{P}(Z_n > \underbrace{Z_{(k)}}_{\text{later}}) \leq 1 - k/n \leq \tau$$

$1 - \frac{\lceil (1 - \tau)n \rceil}{n} \leq \tau$

Corollary 2.6

Let $Z_1, \dots, Z_n \in \mathbb{R}$ be exchangeable. Then $p = \frac{\sum_{i=1}^n \mathbb{1}\{Z_i \geq Z_n\}}{n}$ satisfies $\mathbb{P}(p \leq \tau) \leq \tau$ for all $\tau \in [0, 1]$.

$$\{(X_i, y_i)\}_{i=1}^n$$

X_{n+1} ?

ML, AI, $f(\cdot)$

H_0 : y_{n+1} is outlier

$$R_i = |y_i - f(x_i)|$$

$$p_{n+1} = \frac{1 + \sum_{i=1}^n \mathbb{1}(R_i \geq R_{n+1})}{1 + n}$$

Proving validity of permutation tests

$$T(z_1 \dots z_n) = T(z)$$

$$T(z_{\sigma(1)} \dots z_{\sigma(n)}) = T(z_\sigma)$$

Proof of Theorem 2.4

Step 1: a CDF inequality. First, for any $z \in \mathcal{Z}^n$, write $z_\sigma = (z_{\sigma(1)}, \dots, z_{\sigma(n)})$ to denote the permuted vector for any permutation $\sigma \in \mathcal{S}_n$. We define

$$p = F(v; z) = \frac{\sum_{\sigma \in \mathcal{S}_n} \mathbb{1}\{-T(z_\sigma) \leq v\}}{n!}.$$

We can observe that $F(\cdot; z)$ is the cumulative distribution function (CDF) for the distribution of the quantity $-T(z_\sigma)$, when $\sigma \in \mathcal{S}_n$ is drawn uniformly at random (while z is treated as fixed). We therefore have

$$\mathbb{P}_\sigma(F(-T(z_\sigma); z) \leq \tau) \leq \tau, \quad (2.6)$$

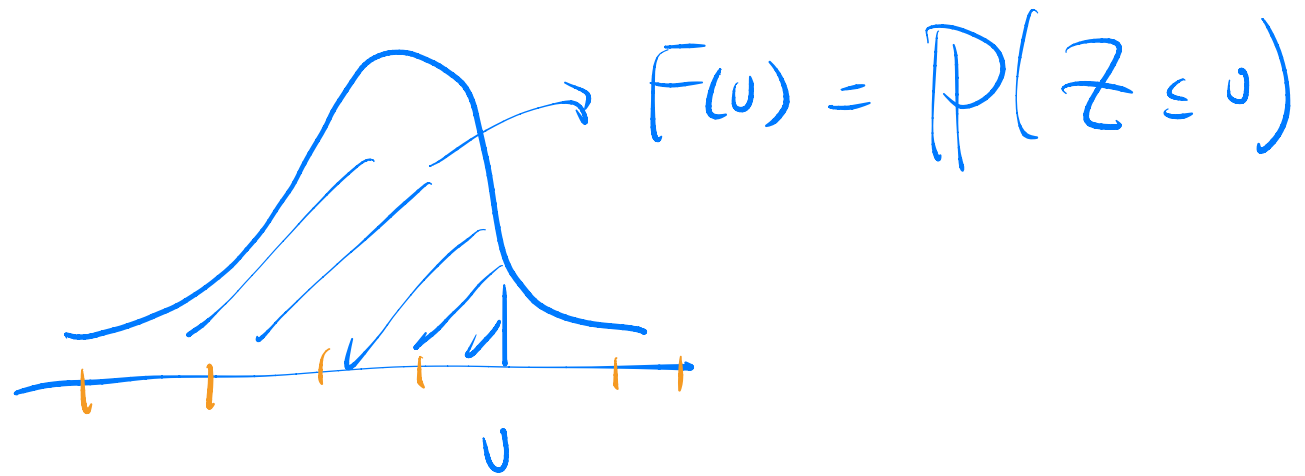
where the probability is taken with respect to $\sigma \in \mathcal{S}_n$ drawn uniformly at random, which is implied directly from the following basic property of CDFs:

If the random variable X has CDF F , then $\mathbb{P}(F(X) \leq \tau) \leq \tau$ for all $\tau \in [0, 1]$. (2.7)

$$\sigma \in \mathcal{S}_n \quad \sigma_1 \cdot \sigma_2 \dots \sigma_{n!}$$

$$\underbrace{-T(z_{\sigma_1}) \dots -T(z_{\sigma_{n!}})}_{z_1 \dots z_n}$$

$$F(v, z) = \frac{\sum_{i=1}^{n!} \mathbb{1}(z_i \leq v)}{n!}$$



$$z_1, \dots, z_n! \sim P$$

Step 2: using exchangeability. Now, we incorporate the exchangeability assumption on $Z = (Z_1, \dots, Z_n)$. For any *fixed* permutation $\sigma \in \mathcal{S}_n$, $Z \stackrel{d}{=} Z_\sigma$ by exchangeability; therefore it also holds that $Z \stackrel{d}{=} Z_\sigma$ when $\sigma \in \mathcal{S}_n$ is drawn uniformly at random (independently of Z).

Next, we observe that the p-value p defined in (2.3) is equal to $p = F(-T(Z); Z)$. Therefore,

$$\mathbb{P}(p \leq \tau) = \mathbb{P}(F(-T(Z); Z) \leq \tau) = \mathbb{P}(F(-T(Z_\sigma); Z_\sigma) \leq \tau) = \mathbb{P}(F(-T(Z_\sigma); Z) \leq \tau),$$

where the last two probabilities are calculated with respect to the distribution of both Z and the randomly drawn σ . Here the second equality holds since $Z \stackrel{d}{=} Z_\sigma$, while the last step holds since $F(v; z) = F(v; z_\sigma)$ for any v, z , and σ , by construction. Finally, we know that $\mathbb{P}(F(-T(Z_\sigma); Z) \leq \tau \mid Z) \leq \tau$, almost surely, by Step 1, which implies that $\mathbb{P}(F(-T(Z_\sigma); Z) \leq \tau) \leq \tau$ by the tower law.